

POLITECNICO DI MILANO



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Final Project SII

Group 7

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1 Original text of the project

Consider a Simplified Insurance Company

ASSETS

- There is a single fund made of equity (80%) and property (20%), $F_t = EQ_t + PR_t$.
- At the beginning ($t = 0$), the value of the fund is equal to the invested premium $F_0 = C_0 = 100,000$.
- Equity features:
 - o Listed in the regulated markets in the EEA.
 - o No dividend yield.
 - o To be simulated with a Risk Neutral GBM (sigma=20%) and a time varying instantaneous rate r .
- Property features:
 - o Listed in the regulated markets in the EEA.
 - o No dividend yield.
 - o To be simulated with a Risk Neutral GBM (sigma=10%) and a time varying instantaneous rate r .

LIABILITIES

- Contract terms:
 - o Whole Life policy.
 - o Benefits:
 - In case of lapse, the beneficiary gets the value of the fund at the time of lapse, with 20 euros of penalties applied.
 - In case of death, the beneficiary gets the maximum between the invested premium and the value of the fund.
 - o Others:
 - Regular Deduction, RD of 2.20%.
 - Commissions to the distribution channels, COMM (or trailing) of 1.40%.
- Model points:
 - o Just 1 model point.
 - o Male with insured aged $x = 60$ at the beginning of the contract.
- Operating assumptions:
 - o Mortality: rates derived from the life table SI2022 (https://demo.istat.it/index_e.php).
 - o Lapse: flat annual rates $lt = 15\%$.
 - o Expenses: constant unitary (i.e., per policy) cost of 50 euros per year, that grows following the inflation pattern.
- Economic assumption:

- o Risk-free: rate r derived from the yield curve (EIOPA IT without VA 31.03.24).
- o Inflation: flat annual rate of 2%.

Other Specifications

- Time horizon for the projection: 50 years.
- In case of outstanding portfolio in $T = 50$, let all the people leave the contract with a massive surrender.
- The interest rates dynamic is deterministic, while the equity and property ones are stochastic.

QUESTIONS

1. Compute Basic Solvency Capital Requirement

Code a Matlab/Python script to compute the Basic Solvency Capital Requirement via Standard Formula and provide comments on the results obtained. The risks to be considered are:

- o Market Interest
- o Market equity
- o Market property
- o Life mortality
- o Life lapse
- o Life cat
- o Expense

2. Split the BEL value into its main PV components

Split the Best Estimate Liability (BEL) value into its main Present Value (PV) components: premiums (= 0), death benefits, lapse benefits, expenses, and commissions.

3. Replicate the calculations in an Excel spreadsheet

Replicate the same calculations in an Excel spreadsheet using a deterministic projection.

- o Do the results differ from 1? If so, what is the reason behind?
- o For the base case only:
 - i. Calculate the Macaulay duration of the liabilities.
 - i. Calculate the sources of profit for the insurance company, deriving its Present Value of Future Profits (PVFP).
 - i. Check the magnitude of leakage by verifying the equation $MVA = BEL + PVFP$ (i.e., $MVA = BEL + PVFP + LEAK$).
 - i. Sense check the PVFP using a proxy calculation, based on the annual profit and the duration of the contract.

4. Open questions

- o What happens to the assets and liabilities when the risk-free rate increases/decreases with a parallel shift of, say, 100bps? Describe the effects for all the BEL components.
- o What happens to the liabilities if the insured age increases? What if there were two model points, one male and one female?

2 Tables of the result

We show the results from the implemented Matlab code (reported in the last part of the report)

Results	MVA	BEL	BoF	d BoF	SCR	Duration
BASE	100000	95834.05	4165.950	0	0	6.243
IR up	100000	95814.501	4185.499	-19.549	0	6.507
IR down	100000	96184.863	3815.137	350.813	350.813	6.120
Equity	68800	67061.243	938.757	3227.193	3227.193	6.350
Property	95000	91221.916	3778.084	387.866	387.866	6.244
Mortality	100000	95864.583	4135.417	30.533	30.533	6.186
Lapse up	100000	97006.898	2993.102	1172.849	1172.849	4.318
Lapse down	100000	93361.752	6638.248	-2472.298	0	10.507
Lapse mass	100000	97435.629	2564.371	1601.579	1601.579	3.882
Catastrophy	100000	95839.217	4160.783	5.167	5.167	6.236
Expenses	100000	95879.527	4120.473	45.476	45.476	6.245
bSCR 4416.456 €						

Tabella 1: Matlab Results

Results	Lapse	Death	Commission	Expenses	Tot
BASE	81259.242	6431.943	7848.396	294.469	95834.05
IRup	81260.297	6431.943	7848.396	273.866	95814.501
IR down	81258.317	6765.743	7848.396	312.406	96184.863
Equity	55898.748	6268.584	5399.441	294.469	67061.243
Property	77197.173	6274.135	7456.140	294.469	91221.916
Mortality	80531.549	7250.403	7790.653	291.978	95864.583
Lapse up	87513.763	3666.696	5621.375	205.064	97006.898
Lapse down	64557.787	15759.288	12549.868	494.809	93361.752
Lapse mass	88740.259	3702.845	4813.912	178.612	97435.629
Catastrophy	81136.560	6569.900	7838.661	294.097	95839.217
Expenses	81259.242	6431.943	7848.396	339.945	95879.527

Tabella 2: Matlab Results

3 Formulas adopted for the calculations

3.1 Base case

We derived rates for the basic scenario from EIOPA IT, then we obtained the forward rates that we will use in the simulation of Equity and Property. To simulate E_t and P_t we have used the Geometric Brownian Motion as follows:

$$E_{t_i} = (1 - RD) \cdot E_{t_{i-1}} \cdot \exp\left[\left(fwd_{t_i} \cdot \frac{\sigma_1^2}{2}\right) \cdot \sigma_1 \cdot n_1\right]$$

where RD is the *regular deduction*, n_1 is a $Normal(0,1)$ and n_2 is a $Normal(0,1)$ independent from n_1 . Then for every time t we take the mean of all the i value at that time and we find the E_t value.

Regarding P_t the simulation is the same as before:

$$P_{t_i} = (1 - RD) \cdot P_{t_{i-1}} \cdot \exp\left[\left(fwd_{t_i} \cdot \frac{\sigma_1^2}{2}\right) \cdot \sigma_1 \cdot n_2\right]$$

Finally, the value of the Fund in every time is the sum of E_t and P_t .

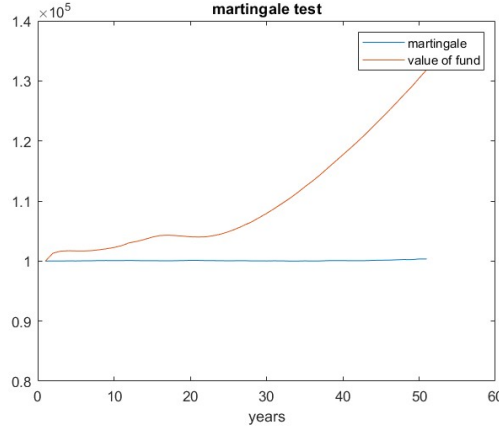
For the implementation of the simulation in Matlab we firstly set two seeds, one for the first Gaussian random variable and one for the second, in order to have the same values every time we run the code and to get independent values.

We also set the number of simulated values to $N = 50$ which is a good trade off between computational time and precision in the result.

In order to check if our simulation was correct we performed the Martingale test.

We simulated as before the Equity and the Property without considering the regular deduction factor and then we discounted back for each year obtaining the conditioning expected value of the fund.

From the plot below we can observe that, while the value of the fund is increasing, the expected value is constant in time and this fact prove that our process is a Martingale.



Then the second task was to compute the liabilities. We firstly obtained the mortality rates from ISTAT, and we used these probabilities to find: the *aggregate survival probability*, named S_t ; the *aggregate lapse probability*, named L_t , that represent the probability to lapse in the t -th year knowing that you are alive in t and you didn't lapse before t ; the *aggregate death probability*, named D_t , that represent the probability to die between the $(t-1)$ -th year and t -th year knowing that you were alive in $t-1$ and you didn't lapse before $t-1$, in this way

$$\begin{cases} S_t = S_{t-1} \cdot (1 - deathProb_t) \\ S(1) = 1 \end{cases}$$

$$L_t = S_t \cdot (1 - lapse)^{t-1} \cdot lapse$$

$$D_t = S_{t-1} \cdot (1 - lapse)^{t-1} \cdot deathProb_t$$

We also found the Expenses **EX** for every year t as

$$EX_t = EX_{t-1} \cdot (1 + infl)$$

Now, having this three probabilities, we can compute the BEL components: *Lapse*, *Death*, *Commission* and *Expenses*

$$Lapse_t = L_t \cdot (F(t) - 20) \cdot discount_t$$

$$Death_t = D_t \cdot max(F(t), C_0) \cdot discount_t$$

$$Commission_t = F(t) \cdot \frac{comm}{(1 - RD)} \cdot S_{t-1} \cdot (1 - lapse)^{t-1} \cdot discount_t$$

$$Expenses_t = EX_t \cdot S_{t-1} \cdot (1 - lapse)^{t-1} \cdot discount_t$$

The liabilities are the sum of this four BEL components, resulting **95834.05** and **BOF=F₀-liabilities=4165.95**. Then we compute ΔBOF as follows:

$$\Delta BOF = BOF_{basic} - BOF$$

In the base case ΔBOF is equal to zero

3.2 IR up

In the case of an upward shift of interest rate, we take new rates again from EIOPA IT. Both assets and liabilities are impacted by this shift, in particular we observe a little decrease in the value of the liabilities, having now a **BEL=95814.501** and a **BOF=4185.499**. So the SCR is 0 since we obtain SCR as:

$$SCR = \max(0, BOF_{base} - BOF_{IRup}) = 0$$

as we notice that the actual BOF is larger than the base case BOF. We can conclude that we are not exposed to the IR up scenario.

3.3 IR down

In the case of downward shift of interest rate, as before we take shifted rates from EIOPA IT. We can state as before that the value of liabilities is different, in particular is larger than base case scenario, and we have a smaller BOF. So in this case SCR is positive and is equal to:

$$SCR = \max(0, BOF_{base} - BOF_{IRdown}) = 350.813$$

We conclude that we are exposed at IR down scenario.

3.4 Equity

In the case of Equity shock, following Solvency II regulation, we have shocked the base case Equity by a 0.39 factor:

$$E_0^{shocked} = E_0 \cdot (1 - 0.39)$$

So we obtained a smaller value of the initial equity, and this have a huge impact in our computation. In particular we observe that the amount of liabilities goes down a lot, having a **BEL=67061.243**. The resulting SCR is way larger than 0:

$$SCR = \max(0, BOF_{base} - BOF_{Equity}) = 3227.193$$

We can expect this result since the BEL components are strictly dependent of the fund value which in this shocked case is decreasing. The only exception are the expenses that are only related to the probability to be in the contract.

3.5 Property

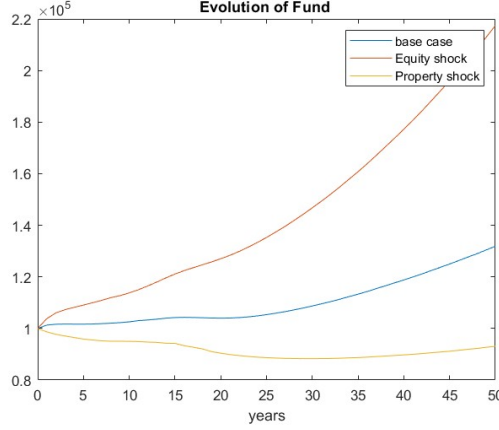
In this case we have to manage property shock. Following Solvency II regulation, we simulated a 25% shocked scenario.

$$P_0^{shocked} = P_0 \cdot (1 - 0.25)$$

Also in this case we have a significant variation in the value of liabilities, but less relevant than the Equity shock, because the fund is composed only by 20% of Property and the remaining part is Equity. In particular, we obtain a positive SCR:

$$SCR = \max(0, BOF_{base} - BOF_{Property}) = 387.866$$

In the following plot we can observe the curves of the shocked scenarios.



3.6 Death

Now we have to analyze what happens in case of an increase of mortality rates. Always following Solvency II, we analyze the scenario of 15% shock up in mortality rates:

$$deathProb^{shocked} = deathProb \cdot (1 + 0.15)$$

In this case we notice a small variation in the value of liabilities, in particular there is a small increase, and consequently a decrease of the BOF, having the following SCR:

$$SCR = \max(0, BOF_{base} - BOF_{Death}) = 30.533$$

This is justified by the fact that the only relevant variation with respect to the base case is in the value of the Death Liabilities, but this factor is less impactful than the other liabilities.

3.7 Lapse up

The following three cases regards the lapse risk, and the capital requirement will be determinate analyzing which of this three cases have the larger SCR.

In this case we have a 50% instantaneous permanent increase of in the assumed option exercise rates of the relevant options in all future years that do not exceed 100%

$$lapse^{up} = \min(1.5 \cdot lapse, 1)$$

We can observe that the firsts and more relevant components of lapse BEL are increasing. This leads to an increasing value of liabilities and a positive SCR:

$$SCR = \max(0, BOF_{base} - BOF_{LapseUp}) = 1172.849$$

3.8 Lapse down

In this case we have an instantaneous permanent decrease of 50% in the assumed option exercise rates of the relevant options in all future years, where the absolute variation do not exceed 20%.

$$lapse^{down} = \max(0.50 \cdot lapse, lapse - 0.20)$$

We observe in this case that the value of liabilities decrease, since the firsts components of lapse BEL are less significant and consequently we have an increase of the BOF, giving us a negative ΔBOF , and a SCR equal to 0:

$$SCR = \max(0, BOF_{base} - BOF_{LapseDown}) = 0$$

3.9 Lapse Mass

In the third case we have an increase of 40% only for the first probability of doing lapse:

$$\begin{cases} lapse^{mass}(1) = lapse + 0.40 \\ lapse^{mass}(t) = lapse \end{cases} \quad \forall t \geq 2$$

In this case we observe the most relevant variation with respect to base case, in particular the liabilities value increase the most, since the first component of lapse BEL is way bigger, giving us the larger SCR:

$$SCR = \max(0, BOF_{base} - BOF_{LapseMass}) = 1601.579$$

We conclude that the Lapse Mass scenario is the one that determine the capital requirement.

3.10 Catastrophy

We analyze now the risk related to an extreme or irregular event. Solvency II impose us to shock by an absolute 1.5 per mille increase in the rate of policyholders dying over the following year:

$$\begin{cases} deathProb^{cat}(1) = deathProb + 0.0015 \\ deathProb^{cat}(t) = deathProb \end{cases} \quad \forall t \geq 2$$

We can notice that there isn't a significant variation of the value of liabilities, indeed the resulting **BEL** is **95839.217**. Moreover we obtain the following SCR:

$$SCR = \max(0, BOF_{base} - BOF_{Catastrophy}) = 5.167$$

This small variation is determined by the fact that the first Death BEL component increased since it is more likely dying within the first year. We can conclude that Catastrophy risk has a small impact in capital requirement.

3.11 Expenses

From Solvency II regulation we can say that the Expense risk arises from the variation in the expenses incurred in servicing insurance and reinsurance contracts.

In particular the expense shock is an increase of 10% in future expenses compared to best estimate anticipations, and increase by 1% per annum of the expense inflation rate compared to anticipations.

$$\begin{cases} EX^{shocked}(0) = EX(0) \cdot (1 + 0.10) \\ EX^{shocked}(t) = EX(0)^{shocked} \cdot (1 + 0.01 + inflation)^t \end{cases} \quad \forall t \geq 2$$

We obtained a small increase of the value of liabilities, due to an increase in the BEL relate to the expenses. So we have a slightly positive SCR:

$$SCR = \max(0, BOF_{base} - BOF_{Expenses}) = 45.476$$

3.12 bSCR

The final objective is to obtain the basic Solvency Capital Requirement.

In particular this is a “sum” of all the individual risks coming from different modules, that are aggregated through linear correlations to take into account the diversification and the mitigation effect.

We use this formula:

$$bSCR = \sqrt{\sum_{i,j} Corr_{i,j} \cdot SCR_i \cdot SCR_j}$$

The two SCR considered in this case are SCR_{life} and SCR_{market} , and we have the following correlation matrix:

$$\begin{bmatrix} 1 & 0.25 \\ 0.25 & 1 \end{bmatrix}$$

So, we have to compute first of all SCR_{life} : we considered the SCR of Life Death, Life Lapse, Life Catastrophy and Life Expenses.

The correlation matrix is the following:

$$\begin{bmatrix} 1 & 0 & 0.25 & 0.25 \\ 0 & 1 & 0.5 & 0.25 \\ 0.25 & 0.5 & 1 & 0.25 \\ 0.25 & 0.25 & 0.25 & 1 \end{bmatrix}$$

The formula that we used is:

$$SCR_{life} = \sqrt{\sum_{i,j} Corr_{i,j} \cdot SCR_i \cdot SCR_j}$$

Then we have to compute SCR_{market} : we considered the SCR of IR shift, Market Equity and Market Property. In particular, since we are exposed to IR down, the correlation matrix will be the following:

$$\begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.75 \\ 0.5 & 0.75 & 1 \end{bmatrix}$$

The formula that we used is:

$$SCR_{market} = \sqrt{\sum_{i,j} Corr_{i,j} \cdot SCR_i \cdot SCR_j}$$

The final result of bSCR using *Matlab* is **bSCR=4416.456**.

We can conclude that bSCR is mostly influenced by Equity of the Market and Lapse of the contract, since they have the most relevant value in the computation of SCR.

4 Deterministic Calculation

4.1 Tables of the result

We show the results from the Excel spreadsheet (delivered together with this report)

Results	MVA	BEL	BoF	d BoF	SCR	dur L
BASE	100000	95795,49	4204,51	0	0	6,24270123
IRup	100000	95775,958	4224,042	-19,5316	0	6,50585296
IR down	100000	96149,636	3850,364	354,1467	354,146694	6,11938541
Equity	68800	67839,248	960,7519	3243,758	3243,758482	6,34977095
Property	95000	91185,827	3814,173	390,3376	390,337582	6,24338096
Mortality	100000	95826,115	4173,885	30,62543	30,62543461	6,18493655
Lapse up	100000	96976,631	3023,369	1181,142	1181,141665	4,31685504
Lapse down	100000	93314,353	6685,647	-2481,14	0	10,5066482
Lapse mass	100000	97411,302	2588,698	1615,812	1615,812028	3,88149526
Catastrophy	100000	95800,702	4199,298	5,212895	5,212895089	6,23540216
Expenses	100000	95840,962	4159,038	45,47222	45,47221757	6,24389545
bSCR 4443,728 €						

Tabella 3: Excel Results

Results	Lapse	Death	Commission	Expenses	Tot
BASE	81227,12939	6428,626	294,4506162	7845,283646	95795,4896
IRup	81228,18276	6428,626	273,86569	7845,283646	95775,95805
IR down	81226,2	6765,743	312,405812	7845,283646	96149,6363
Equity	55879,29	6267,951	294,45062	5397,555148	67839,24808
Property	77164,97583	6273,381	294,45062	7453,019464	91185,82718
Mortality	80499,9	7246,677	291,960713	7787,578813	95826,11504
Lapse up	87486,73	3665,209	205,05699	5619,634303	96976,63127
Lapse down	64525,84	15750,12	494,758815	12543,63848	93314,35333
Lapse mass	88719,42	3701,064	178,6034759	4812,208989	97411,3
Catastrophy	81104,5	6566,575	294,0789495	7835,552783	95800,7
Expenses	81227,13	6428,626	339,9228337	7845,283646	95840,96182

Tabella 4: Excel Results

4.2 Comments on the results

Regarding the deterministic calculations done via Excel are overall similar to the Matlab one.

We have however some differences:

The main difference is related to the computation of the fund. In Matlab, as illustrated before, we use a Geometric Brownian Motion to simulate the value. In this case instead, we perform a deterministic calculation: In particular,

$$E_i = E_{i-1} \cdot (1 - RD) \cdot \exp[(t_i - t_{i-1}) \cdot fwd_{t_i}]$$

and

$$P_i = P_{i-1} \cdot (1 - RD) \cdot \exp[(t_i - t_{i-1}) \cdot fwd_{t_i}]$$

so the value of the fund F_i is

$$F_i = (P_{i-1} + E_{i-1}) \cdot (1 - RD) \cdot \exp[(t_i - t_{i-1}) \cdot fwd_{t_i}]$$

The final result are quite similar as we expect, and all the other consideration are the same as the case in *Matlab*

4.3 Macaulay duration

To compute the Macaulay Duration of the liabilities we used the following formula:

$$MacD = \frac{\sum_{i=1}^T t_i \cdot Liabilities_i \cdot discounts_i}{totalLiabilities}$$

In the base case, the result is **6,242701**.

As we can observe in the Tables of the Result, there are only three cases of risk where the duration is significantly different.

This three cases are: Lapse up, Lapse down and Laps mass. This is justified by the fact that the element that mostly characterize the liabilities is the Lapse BEL and as we can see in the formula the liabilities are strongly related to the Macaulay Duration.

So, having a shift of the Lapse probability changes a lot the Duration. In particular, in case of Lapse up and Mass, we have a small duration, while in case of Lapse down, the duration significantly increases.

4.4 Source of profit (PVFP)

Now we want to calculate the sources of profit for the insurance company, deriving its PVFP. To compute the PVFP first of all we obtain the gain, as

$$Gain_t = \frac{(RD - comm) \cdot F_t \cdot P_{nlnd}}{(1 - RD)}$$

where

$$P_{nlnd} = (1 - probDeath)(1 - probLapse)$$

Then, to obtain PVFP:

$$PVFP = \sum_{t=1}^T Gain_t \cdot discount_t = 4483.0192$$

4.5 Magnitude of leakage

Then we check the magnitude of leakage by verifying that $MVA \sim BEL + PVFP$, and the LEAK represent the "error", i.e. $MVA = BEL + PVFP + LEAK$.

So, computing LEAK is trivial because we only have to invert the previous formula:

$$LEAK = MVA - BEL - PVFP = -278.5088$$

where MVA=100000 and PVFP is that one calculate previously.

We can state that the LEAK is approximately 0.3% of the fund, that is an acceptable result.

4.6 Proxy calculation for PVFP

The last thing is to sense check the PVFP using a proxy calculation, based on the annual profit and the duration of the contract. To perform this calculation we used the following idea:

$$PVFP_{proxy} = (RD - comm) \cdot C_0 \cdot MacD = 4994.1609$$

Then we also repeat the check of magnitude of the leakage, obtaining

$$LEAK_{proxy} = MVA - BEL - PVFP = -789.651$$

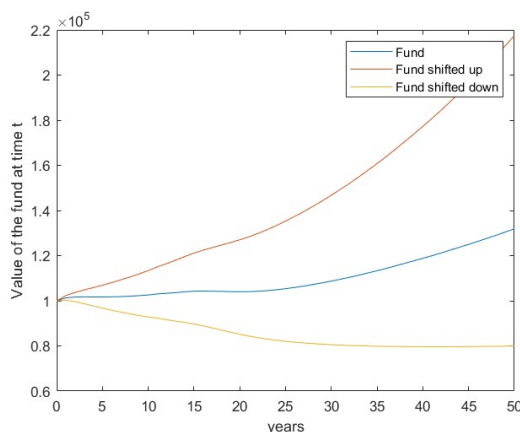
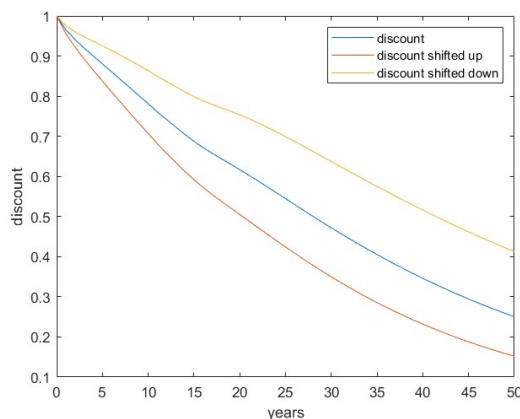
We can observe that in this case the LEAK is larger in absolute value with respect to the previous calculation, but it is useful because allow us to get an estimation of the order of magnitude in a simpler way.

5 Open questions

5.1 Question 1

In the first question we were asked to observe what happens to the asset and liabilities when the risk-free rate increases/decreases with a parallel shift of 100bps and to describe the effects for all the BEL components.

We can observe in the following plots that if we have a parallel shift upward of 100bps the fund will go up, while the discount curve will go down. Viceversa, if we have a parallel shift downward the fund will go down and the discount curve will go up.



Regarding BEL we have in increasing rate case:

- The Lapse BEL is almost equal to base case, indeed we have at the same time an increase of the fund and a decrease of the discount that compensate the first variation. Moreover the most significant cash flows are paid in the first years and so they are not so influenced.
- The Death BEL is equal to base case, we can justify this by the fact that when we take the maximum between the value of the fund and the invested premium, as we can see in the plot, we will take always the value of the fund that is strictly bigger. So, when we multiply this values by the respective discount factors, that are smaller than the base case, we have a compensation of the final result.

- The Commission BEL remain exactly equal to base case, and we expect this because as before at the same time an increase of the fund and a decrease of the discount that compensate the first variation.
- The Expenses BEL decrease, indeed they are influenced only by discount factors, which when the rates goes up are smaller.

In the decreasing rate case we have:

- The Lapse BEL is equal to base case and we justify this as the increasing case.
- The Death BEL increase since when we take the maximum between the value of the fund and the invested premium, that in this case is always equal to the invested premium. Moreover also the discount factor are bigger and so the resulting death BEL is bigger.
- The Commission BEL also remain equal to base case as explained before.
- The Expenses BEL increase, because in this case the discount factors are increasing.

The following table resumes the final bSCR in the three cases:

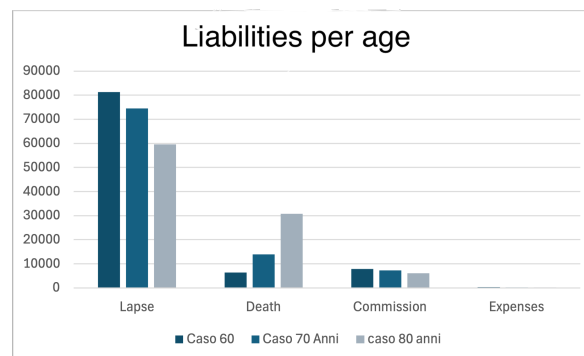
bSCR base case	bSCR rates up	bSCR rates down
4416.45 €	3856.41 €	4388.04 €

5.2 Question 2

In this final question we were asked to investigate what happens to liabilities if the insured age increase and if we consider a two model point with one male and one female.

We started considering the insured age to 70y and we compute the same computations using the implemented Matlab code changing only the mortality rate considering a time horizon of 40y from 70y to 110y. For the sake of analysis we repeated also setting the insured age to 80y. As we can see in the chart below if the insured age becomes higher we have that:

- The Lapse BEL decreases since the probability to get these benefits is related to the survival probability which is, of course, lower year by year.
- The Death BEL increase since the probability of dying and consequently to pay them is increasing.
- The Commission BEL decreases since the average of the number of years that the policy holder will be in the contract is lower
- The Expenses BEL decreases for the same reason of the Commission Bel.



We also computed for this two new cases the bSCR as reported in the following table:

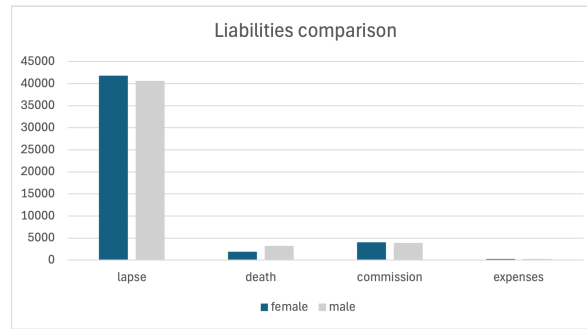
bSCR 60y	bSCR 70y	bSCR 80y
4416.45 €	6688.48 €	12001.96 €

We can notice that the bSCR is increasing due to the fact that the probability that the Insurance Company will pay the policy holder in the next few years is increasing and so, to handle this future cash flow, the capital to set aside is greater.

In the second case, if we have a two model point, with one male and one female, we have first of all obtained the new probability of death related to female population from ISTAT.

Then we have changed all the initial values, dividing by an half, in this way we can analyze the model with both female and male, with the same initial value. So, for example, we divided the initial value of the fund, F_0 in 50k for male population and 50k for female and so on.

Here we report the plot of the new result obtained:



We can observe that the Death BEL is the one that varies the most, since we know that in average the female population has a bigger life expectation, so the male Death BEL is way bigger.

The other three value are quite similar, but in all this cases the value of female Liabilities are a little bit bigger since, again, they live more years.

Because of this factor, we have a change of the bSCR, in particular, the **male bSCR** results **2186.29 €**, while the **female bSCR** is smaller, more precisely **bSCR=1804.97 €**.

6 Matlab code

In this final section we reported the Matlab code. It is structured with the following 3 blocks:

- main script named 'run group 7' where there are the computations for every different scenario
- a function named 'simulation' which simulate the values of the Fund during the years. This function is called in the main script when it's needed to simulate
- a function named 'computeliabilities' which compute the differents components and quantities related to the liabilities. Also this function is called in the main script when we needed to compute the liabilities for the different scenario.

Listing 1: Run Group 7

```
%% INSURANCE PROJECT GROUP 7
% Falconi Neri
% Frigerio Emanuele
% Garagiola Stefano
% Giboni Lorenzo

clear all
clc

% setting datas
F0=100000; % value of the fund
Eq0=80000; % value of equity in t=0
Pr0=20000; % value of the property in t=0
C0=100000; % invested premium in t=0
sigma1=0.2; % sigma for equity
sigma2=0.1; % sigma for property
t=1:50; % vector of years
years=0:50; % years
RD=0.022; % regular deduction
comm=0.014; % commission
x=60; % age insured person
lapse=0.15*ones(1,50); % probability of lapse
infl=0.02; % inflation
T=50; % time horizon

% Rates for the basic scenario taken from EIOPA
rates = xlsread('EIOPA_RFR_20240331_Term_Structures.xls', 'RFR_spot_no_VA', 'C11:C60')'-100*1e-4;

% parallel shift
rates_shift_up=rates + 100*1e-4;
rates_shift_down=rates - 100*1e-4;

% Rates shocked up taken from EIOPA
rates_up = xlsread('EIOPA_RFR_20240331_Term_Structures.xls', 'Spot_NO_VA_shock_UP', 'C11:C60')';

% Rates shocked down taken from EIOPA
rates_down = xlsread('EIOPA_RFR_20240331_Term_Structures.xls', 'Spot_NO_VA_shock_DOWN', 'C11:C60')';

% death probability
death_prob= xlsread('Tavole-di-mortalit -della-popolazione-residente.xls', 'D63:D113')/1000;
```

```

% initialization
surv_prob_aggregate(1)=1;
expenses(1)=50;
death_prob_aggregate(1)=0;
% computing probability needed and expenses
for j=1:T
    surv_prob_aggregate(j+1)=surv_prob_aggregate(j)*(1-death_prob(j));
    lapse_prob_aggregate(j)=surv_prob_aggregate(j+1)*(1-lapse(j))^(j-1)*lapse(j);
    death_prob_aggregate(j+1)=surv_prob_aggregate(j)*(1-lapse(j))^(j-1)*death_prob(j);

    expenses(j+1)=expenses(j)*(1+infl); % compute every year expenses
end

% random simulation settings
rng(145);
N=1000000;
n1=randn(T,N);
rng(56)
n2=randn(T,N);

% compute discount and forward rates base
discount=zeros(T+1,1);
discount(1)=1;

discount(2:end)=exp(-rates.*t);
fwd_discounts=discount(2:end)./discount(1:end-1);
fwd_rates=-log(fwd_discounts);

%% base case
F=simulation(fwd_rates,sigma1,sigma2,n1,n2,N,RD,Eq0,Pr0,T);

[BEL_components_base,liabilities_tot_base,BOF_base,duration_base]=
compute_liabilities(lapse_prob_aggregate,death_prob_aggregate,surv_prob_aggregate,lapse,
expenses,comm,RD,F,F0,C0,discount(2:end),T);
delta_BOF_base=0;
SCR_base=max(delta_BOF_base,0);

%% Martingale test
F_mg=simulation(fwd_rates,sigma1,sigma2,n1,n2,N,0,Eq0,Pr0,T);
figure()
plot([1:51],F_mg.*discount)
hold on
plot([1:51],F)
ylim([0.8*1e5, 1.4*1e5])
xlabel('years')
title('martingale test')
legend('martingale','value of fund')

%% rates up
% compute discount up and forward rates up
discount_up=zeros(T+1,1);
discount_up(1)=1;
discount_up(2:end)=exp(-rates_up.*t);

```

```

fwd_discounts_up=discount_up(2:end)./discount_up(1:end-1);
fwd_rates_up=-log(fwd_discounts_up);

F_up=simulation(fwd_rates_up,sigma1,sigma2,n1,n2,N,RD,Eq0,Pr0,T);

[BEL_components_up,liabilities_tot_up,BOF_up,duration_up]=
compute_liabilities(lapse_prob_aggregate,death_prob_aggregate,surv_prob_aggregate,lapse,
expenses,comm,RD,F_up,F0,C0,discount_up(2:end),T);
delta_BOF_up=BOF_base - BOF_up;
SCR_up=max(delta_BOF_up,0);

%% rates down
% compute discount down and forward rates down
discount_down=zeros(T+1,1);
discount_down(1)=1;
discount_down(2:end)=exp(-rates_down.*t);
fwd_discounts_down=discount_down(2:end)./discount_down(1:end-1);
fwd_rates_down=-log(fwd_discounts_down);

F_down=simulation(fwd_rates_down,sigma1,sigma2,n1,n2,N,RD,Eq0,Pr0,T);

[BEL_components_down,liabilities_tot_down,BOF_down,duration_down]=
compute_liabilities(lapse_prob_aggregate,death_prob_aggregate,surv_prob_aggregate,lapse
,expenses,comm,RD,F_down,F0,C0,discount_down(2:end),T);
delta_BOF_down=BOF_base - BOF_down;
SCR_down=max(delta_BOF_down,0);

%% equity shocked
Eq0_shocked=Eq0*(1-0.39);
F0_shocked=Eq0_shocked + Pr0;

F_equity_shocked=simulation(fwd_rates,sigma1,sigma2,n1,n2,N,RD,Eq0_shocked,Pr0,T);

[BEL_components_equity,liabilities_tot_equity,BOF_equity,duration_equity]=
compute_liabilities(lapse_prob_aggregate,death_prob_aggregate,surv_prob_aggregate,lapse,expenses,
comm,RD,F_equity_shocked,F0_shocked,C0,discount(2:end),T);
delta_BOF_equity=BOF_base - BOF_equity;
SCR_equity=max(delta_BOF_equity,0);

%% property shocked
Pr0_shocked=Pr0*(1-0.25);
F0_property_shocked=Eq0 + Pr0_shocked;

F_property_shocked=simulation(fwd_rates,sigma1,sigma2,n1,n2,N,RD,Eq0,Pr0_shocked,T);

[BEL_components_property,liabilities_tot_property,BOF_property,duration_property]=
compute_liabilities(lapse_prob_aggregate,death_prob_aggregate,surv_prob_aggregate,lapse,expenses,
comm,RD,F_property_shocked,F0_property_shocked,C0,discount(2:end),T);
delta_BOF_property=BOF_base - BOF_property;
SCR_property=max(delta_BOF_property,0);

%% plot evolution of the fund
figure()
plot([0:50],F)

```

```

hold on
plot([0:50],F_up)
plot([0:50],F_down)
ylim([0.8*1e5, 2.2*1e5])
xlabel('years')
title('Evolution of Fund')
legend('base-case','Equity-shock','Property-shock')

%% mortality shocked
% compute new probability
death_prob_shocked=death_prob * 1.15;
surv_prob_aggregate_mortality(1)=1;
death_prob_aggregate_mortality(1)=0;
for j=1:T
    surv_prob_aggregate_mortality(j+1)=surv_prob_aggregate_mortality(j)*(1-death_prob_shocked(j));
    lapse_prob_aggregate_mortality(j)=surv_prob_aggregate_mortality(j+1)*(1-lapse(j))^(j-1)*lapse(j);
    death_prob_aggregate_mortality(j+1)=surv_prob_aggregate_mortality(j)*(1-lapse(j))^(j-1)
    *death_prob_shocked(j);    %(1-lapse(j-1))
end

[BEL_components_mortality,liabilities_tot_mortality,BOF_mortality,duration_mortality]=
compute_liabilities(lapse_prob_aggregate_mortality,death_prob_aggregate_mortality,
surv_prob_aggregate_mortality,lapse,expenses,
comm,RD,F,F0,C0,discount(2:end),T);
delta_BOF_mortality=BOF_base - BOF_mortality;
SCR_mortality=max(delta_BOF_mortality,0);

%% lapse up
lapse_up=min(lapse*1.5,1);

death_prob_aggregate_lup(1)=0;

% computing new probability needed
for j=1:T
    lapse_prob_aggregate_lup(j)=surv_prob_aggregate(j+1)*(1-lapse_up(j))^(j-1)*lapse_up(j);
    death_prob_aggregate_lup(j+1)=surv_prob_aggregate(j)*(1-lapse_up(j))^(j-1)*death_prob(j);
end

[BEL_components_lapse_up,liabilities_tot_lapse_up,BOF_lapse_up,duration_lapse_up]=
compute_liabilities(lapse_prob_aggregate_lup,death_prob_aggregate_lup,surv_prob_aggregate,lapse_up,
expenses,comm,RD,F,F0,C0,discount(2:end),T);
delta_BOF_lapse_up=BOF_base - BOF_lapse_up;
SCR_lapse_up=max(delta_BOF_lapse_up,0);

%% lapse down
lapse_down=max(lapse*0.5,lapse-0.2);

death_prob_aggregate_ldown(1)=0;

% computing new probability needed
for j=1:T
    lapse_prob_aggregate_ldown(j)=surv_prob_aggregate(j+1)*(1-lapse_down(j))^(j-1)*lapse_down(j);
    death_prob_aggregate_ldown(j+1)=surv_prob_aggregate(j)*(1-lapse_down(j))^(j-1)*death_prob(j);

```

end

```
[BEL_components_lapse_down, liabilities_tot_lapse_down, BOF_lapse_down, duration_lapse_down]=  
compute_liabilities(lapse_prob_aggregate_ldown, death_prob_aggregate_ldown,   
surv_prob_aggregate, lapse_down, expenses,   
comm, RD, F, F0, C0, discount(2:end), T);  
delta_BOF_lapse_down=BOF_base - BOF_lapse_down;  
SCR_lapse_down=max(delta_BOF_lapse_down, 0);
```

%% lapse mass

```
lapse_mass_shocked=lapse(1) + 0.4; % the effect is only on the first year  
lapse_mass=[lapse_mass_shocked, lapse(2:end)];  
death_prob_aggregate_lapse_mass(1)=0;
```

% computing new probability

```
for j=1:T  
    if j==1  
        lapse_prob_aggregate_lapse_mass(j)=surv_prob_aggregate(j+1)*lapse_mass(j);  
        death_prob_aggregate_lapse_mass(j+1)=surv_prob_aggregate(j)*death_prob(j);    %(1-lapse(j-1))  
    else  
        lapse_prob_aggregate_lapse_mass(j)=surv_prob_aggregate(j+1)*(1-lapse_mass(1))  
        *(1-lapse_mass(j))^(j-2)*lapse_mass(j);  
        death_prob_aggregate_lapse_mass(j+1)=surv_prob_aggregate(j)*(1-lapse_mass(1))  
        *(1-lapse(j))^(j-2)*death_prob(j);  
    end  
end
```

```
[BEL_components_lapse_mass, liabilities_tot_lapse_mass, BOF_lapse_mass, duration_lapse_mass]=  
compute_liabilities(lapse_prob_aggregate_lapse_mass,   
death_prob_aggregate_lapse_mass, surv_prob_aggregate, lapse_mass, expenses,   
comm, RD, F, F0, C0, discount(2:end), T);  
delta_BOF_lapse_mass=BOF_base - BOF_lapse_mass;  
SCR_lapse_mass=max(delta_BOF_lapse_mass, 0);
```

%% expenses

```
expenses_shocked(1)=50*1.1;  
inflation_shocked=infl + 0.01;  
for j=1:50  
    expenses_shocked(j+1)=expenses_shocked(j)*(1+inflation_shocked); % compute every year expenses  
end
```

```
[BEL_components_expenses, liabilities_tot_expenses, BOF_expenses, duration_expenses]=  
compute_liabilities(lapse_prob_aggregate, death_prob_aggregate,   
surv_prob_aggregate, lapse,   
expenses_shocked, comm, RD, F, F0, C0, discount(2:end), T);  
delta_BOF_expenses=BOF_base - BOF_expenses;  
SCR_expenses=max(delta_BOF_expenses, 0);
```

%% catastrophe

```
death_prob_catastrophy=[death_prob(1)+1.5/1000; death_prob(2:end)];
```

```

surv_prob_aggregate_catastrophy(1)=1;
death_prob_aggregate_catastrophy(1)=0;

% computing probability needed
for j=1:T
    surv_prob_aggregate_catastrophy(j+1)=surv_prob_aggregate_catastrophy(j)*
    (1-death_prob_aggregate_catastrophy(j));

    lapse_prob_aggregate_catastrophy(j)=surv_prob_aggregate_catastrophy(j+1)
    *(1-lapse_prob_aggregate_catastrophy(j));

    death_prob_aggregate_catastrophy(j+1)=surv_prob_aggregate_catastrophy(j)
    *(1-lapse_prob_aggregate_catastrophy(j));

end

[BEL_components_catatsrophy, liabilities_tot_catastrophy, BOF_catastrophy, duration_catastrophy]=
compute_liabilities(lapse_prob_aggregate_catastrophy,
death_prob_aggregate_catastrophy, surv_prob_aggregate_catastrophy,
lapse, expenses, comm, RD, F, F0, C0, discount(2:end), T);
delta_BOF_catastrophy=BOF_base - BOF_catastrophy;
SCR_catastrophy=max(delta_BOF_catastrophy, 0);

%% BSCR
SCR_IR=max(SCR_down, SCR_up);
SCR_MKT=[SCR_IR, SCR_equity, SCR_property];

% correlation matrix market
if SCR_up>SCR_down
    MKT_corr_matrix = [1 0 0; 0 1 0.75; 0 0.75 1];
else
    MKT_corr_matrix = [1 0.5 0.5; 0.5 1 0.75; 0.5 0.75 1];
end

BSCR_MKT=sqrt(SCR_MKT*MKT_corr_matrix*SCR_MKT');

SCR_lapse=max(SCR_lapse_up, max(SCR_lapse_down, SCR_lapse_mass));
SCR_LIFE=[SCR_mortality, SCR_lapse, SCR_expenses, SCR_catastrophy];

% correlation matrix life
LIFE_corr_matrix=[1 0 0.25 0.25; 0 1 0.5 0.25; 0.25 0.5 1 0.25; 0.25 0.25 0.25 1];

BSCR_LIFE=sqrt(SCR_LIFE*LIFE_corr_matrix*SCR_LIFE');

% correlation BSCR
BSCR_correlation_matrix=[1 0.25; 0.25 1];

BSCR_partial=[BSCR_MKT, BSCR_LIFE];

% computation for BSCR
BSCR=sqrt(BSCR_partial*BSCR_correlation_matrix*BSCR_partial')

```

Listing 2: liabilities computation function

```

function [BEL_components,liabilities_tot ,BOF,Duration]=compute_liabilities (lapse_prob_aggregate ,
death_prob_aggregate ,surv_prob_aggregate ,lapse ,expenses ,comm,RD,F,F0,C0,discount ,T)
%
% function to compute BEL components, total liabilities , BOF and duration
%
% INPUT
%
% lapse_prob_aggregate:    vector of probaility of doing lapse
% death_prob_aggregate:    vector of probaility of dying
% surv_prob_aggregate:     vector of probaility of surviving
% lapse:                  vector of lapse rate
% expenses:               vector of expenses
% comm:                   commission rate
% RD:                     regular deduction
% F:                      vector of the values of the fund
% F0:                    value of the fund i t=0
% C0:                    invested premium
% discount:              discount factors
% T:                     time horizon
%
% OUTPUT
% BEL_components:         vector of BEL components (lapse , death ,comm,exp)
% liabilities_tot:        total liabilities
% BOF:                   BOF
% Duration:              mmacaulay duration
% for loop to compute liabilities
for j=1:T
    lapse_ben(j)=lapse_prob_aggregate(j)*(F(j+1)-20);
    death_ben(j)=death_prob_aggregate(j+1)*max(F(j+1),C0);

    % condition for to detect the change in lapse rate
    if j==1
        commission(j)=F(j+1)*comm/(1-RD)*surv_prob_aggregate(j);    lapse(j));
        expenses_liab(j)=expenses(j)*surv_prob_aggregate(j);
    else
        commission(j)=F(j+1)*comm/(1-RD)*surv_prob_aggregate(j)*(1-lapse(1))
        *(1-lapse(j))^(j-2);
        expenses_liab(j)=expenses(j)*surv_prob_aggregate(j)*(1-lapse(1))
        *(1-lapse(j))^(j-2);
    end
    partial_sum(j)=lapse_ben(j)+death_ben(j)+commission(j)+expenses_liab(j);

end
% computations of the output

BEL_components=[lapse_ben*discount ; death_ben*discount ; commission*discount ;
expenses_liab*discount ];
liabilities_tot=sum(BEL_components);
BOF=F0-liabilities_tot ;
t=(1:50)';
Duration=partial_sum*t/sum(partial_sum);

end

```

Listing 3: simulation function

```

function F=simulation(fwd_rates ,sigma_equity ,sigma_property ,n1 ,n2 ,N,RD,Equity_in ,Property_in ,T)
%
% This function compute and return the simulated values of the fund for
% every year
%
% INPUT
%
% fwd_rates:          forward rates
% sigma_equity:       volatility for the equity
% sigma_property:     volatility for the property
% n1:                 matrix 50xN of guassian random values
% n2:                 matrix 50xN of guassian random values
% N:                  number of Monte Carlo simulation
% RD:                 regular deduction
% Equity_in:          initial value of the equity
% Property_in:        initial value of the equity
% T:                  time horizon
%
% OUTPUT
%
% F:                  simulated values of the fund for
%                     every year

E(1,:)=Equity_in*ones(1,N); % initialize matrixes
P(1,:)=Property_in*ones(1,N);
% for loop in order to simulate
for i=2:T+1
    E(i,:)=(1-RD)*E(i-1,:).*exp((fwd_rates(i-1)-0.5*sigma_equity^2)+sigma_equity.*n1(i-1,:));

    P(i,:)=(1-RD)*P(i-1,:).*exp((fwd_rates(i-1)-0.5*sigma_property^2)+sigma_property.*n2(i-1,:));
end

Equity=mean(E,2);
Property=mean(P,2);

F=Equity+Property; % value of the fund
end

```